Name:_____

TOPIC TEST 1

Chapter 6 Applications of integration

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Five multiple-choice questions (5 marks).
- Part B: Six short-answer questions (27 marks).
- Part C: Two analysis questions (8 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 $\int_0^2 [5f(x) + 3]dx =$

A $\int_0^2 f(x)dx + 3x$

B $5\int_0^2 f(x)dx + 3x$

C $5f'(x) + 3$

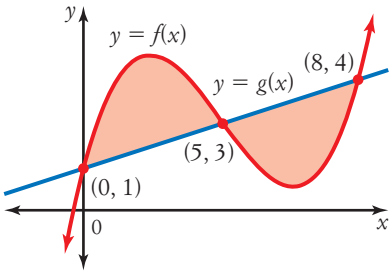
D $5\int_0^2 f(x)dx + 6$

E $5\int_0^2 f(x)dx + x$

[1 mark]

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2 The shaded area below can be written as:



A $\int_0^5 f(x) - g(x)dx - \int_5^8 g(x) - f(x)dx$

B $\int_0^8 f(x) - g(x)dx$

C $\int_0^5 f(x) - g(x)dx + \int_5^8 g(x) - f(x)dx$

D $\int_1^3 f(x) - g(x)dx + \int_3^4 g(x) - f(x)dx$

E $\int_1^3 f(x) - g(x)dx - \int_3^4 g(x) - f(x)dx$

[1 mark]

3 $\int_0^4 (6\sqrt{x} - x)dx =$

A 64

B 48

C 40

D 24

E 12

[1 mark]

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4 If the derivative of $e^{x^2 - 6x}$ is $2(x - 3)\ e^{x^2 - 6x}$, then the antiderivative of $(x - 3)\,e^{x^2 - 6x}$ is:

A $2e^{x^2 - 6x} + c$

B $\frac{1}{2}(x - 3)\ e^{x^2 - 6x}$

C $e^{x^2 - 6x} + c$

D $2(x - 3)\ e^{x^2 - 6x}$

E $\frac{1}{2}e^{x^2 - 6x} + c$

[1 mark]

5 Water flows out of a container at the rate
$R'(t) = 3e^{-0.2t}$ (L/min)
where t is in minutes.
What is the total number of litres (to the nearest litre) that flowed out of the container in the first 5 minutes?

A 12

B 9

C 20

D 15

E 95

[1 mark]

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Part B

Short-answer questions

Six short-answer questions
31 marks
Show your working where appropriate.

1 Find each indefinite integral.

a $\int (x^5 - 6)dx$

b $\int (3 + 2x)^2 \, dx$

c $\int \frac{6x^2 + 2x + 1}{\sqrt{x}} dx$

[4 marks]

2 Evaluate the following:

a $\int_{-1}^{3} (6x^2 + 4x - 1)dx$

b $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} 6 \cos(3x)dx$

c $\int_{2}^{5} \frac{dx}{(x + 3)^2}$

[6 marks]

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3 Find y in terms of x if $\frac{dy}{dx} = 8x - 7$ and $y = 13$ when $x = -1$.

[3 marks]

4 Find the area under the curve $y = x^2 - 4x - 12$ from $x = -1$ to $x = 4$.

[5 marks]

5 Find the area enclosed between the curves $y = x^2$, $y = (x - 4)^2$ and the x-axis.

[6 marks]

6 Fluid flows into a storage tank at the rate

$F'(t) = 100e^{0.2t}$ litres/h

where t is measured in hours.

How much fluid will flow into the tank in the first 3 hours?

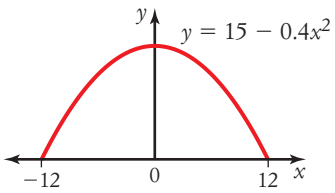
[3 marks]

Part C

Analysis questions

Two analysis questions
8 marks
Show your working where appropriate.

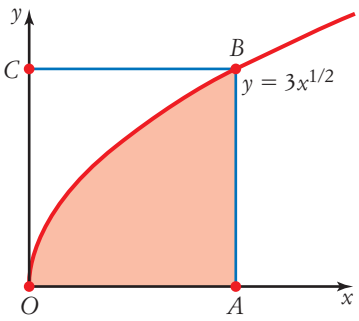
1 A large warehouse has a cross-section as shown below. The roof has the shape of a function $y = 15 - 0.4x^2$ $(-12 \leq x \leq 12)$.



If the warehouse is 80 m long, find its volume.

[3 marks]

2 This diagram shows the function $y = 3x^{\frac{1}{2}}$. B lies on the curve and O is the origin.



Prove that the shaded area is two-thirds of the area of the rectangle $OABC$.

[5 marks]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $\int_0^2 [5f(x) + 3]dx = 5\int_0^2 f(x)dx + \int_0^2 3dx$

$= 5\int_0^2 f(x)dx + [3x]_0^2 = 5\int_0^2 f(x)dx + 6$

$\therefore$ **D** **[1 mark]**

2 Area between $x = 0$ and $x = 5$ is $\int_0^5 f(x) - g(x)dx$

Area between $x = 5$ and $x = 8$ is $\int_5^8 g(x) - f(x)dx$ total area $= \int_0^5 f(x) - g(x)dx + \int_5^8 g(x) - f(x)dx$

$\therefore$ **C** **[1 mark]**

3 $\int_0^4 (6\sqrt{x} - x)dx = \int_0^4 (6x^{\frac{1}{2}} - x)dx$

$$= \left[\frac{6 \times 2x^{\frac{3}{2}}}{3} - \frac{x^2}{2}\right]_0^4$$

$$= \left[4x^{\frac{3}{2}} - \frac{x^2}{2}\right]_0^4$$

$$= \left(4 \times 4^{\frac{3}{2}} - \frac{4^2}{2}\right) - (0 - 0)$$

$$= 4 \times 2^3 - 8$$

$$= 24$$

$\therefore$ **D** **[1 mark]**

4 $\frac{d}{dx}e^{x^2-6x} = 2(x - 3)\ e^{x^2-6x}$

So $\int 2(x - 3)e^{x^2-6x}\,dx = e^{x^2-6x} + c$

So $\int (x - 3)e^{x^2-6x}\,dx = \frac{1}{2}\int (x - 3)e^{x^2-6x}\,dx$

$$= \frac{1}{2}e^{x^2-6x} + c$$

$\therefore$ **E** **[1 mark]**

5 Total change $= \int_a^b R'(t)dt$

$$= \int_0^5 3e^{-0.2t}\,dt$$

$$= \left[\frac{3e^{-0.2t}}{-0.2}\right]_0^5$$

$$= \left[-15e^{-0.2t}\right]_0^5$$

$$= -15e^{-1} + 15e^0$$

$$= 9.4818\ldots$$

$\therefore$ **B** **[1 mark]**

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Part B – Short-answer questions

1 a $\int (x^5 - 6)dx = \int x^5\,dx - \int 6dx$

$= \frac{x^6}{6} - 6x + c$ **[1 mark]**

b $\int (3 + 2x)^2\,dx = \int (9 + 12x + 4x^2)dx$

$= 9x + \frac{12x^2}{2} + \frac{4x^3}{3} + c$

$= 9x + 6x^2 + \frac{4x^3}{3} + c$ **[1 mark]**

c $\int \frac{6x^2 + 2x + 1}{\sqrt{x}}dx = \int (6x^2 + 2x + 1)x^{-\frac{1}{2}}dx$

$= \int (6x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + x^{-\frac{1}{2}})dx$ **[1 mark]**

$= \frac{6x^{\frac{5}{2}}}{\frac{5}{2}} + \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + c$

$= \frac{2 \times 6x^{\frac{5}{2}}}{5} + \frac{2 \times 2x^{\frac{3}{2}}}{3} + 2x^{\frac{1}{2}} + c$

$= \frac{12x^2\sqrt{x}}{5} + \frac{4x\sqrt{x}}{3} + 2\sqrt{x} + c$ **[1 mark]**

2 a $\int_{-1}^{3} (6x^2 + 4x - 1)dx = \left[\frac{6x^3}{3} + \frac{4x^2}{2} - x\right]_{-1}^{3}$

$= \left[2x^3 + 2x^2 - x\right]_{-1}^{3}$ **[1 mark]**

$= \left[2 \times 3^3 + 2 \times 3^2 - 3\right] - \left[2 \times (-1)^3 + 2 \times (-1)^2 + 1\right]$

$= 69 - 1$

$= 68$ **[1 mark]**

b $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} 6\cos(3x)dx = \left[\frac{6\sin(3x)}{3}\right]_{-\frac{\pi}{3}}^{\frac{\pi}{3}}$

$= \left[2\sin(3x)\right]_{-\frac{\pi}{3}}^{\frac{\pi}{3}}$ **[1 mark]**

$= 2\sin(\pi) - 2\sin(-\pi)$

$= 0 - 0$

$= 0$ **[1 mark]**

c $\int_2^5 \frac{dx}{(x+3)^2} = \int_2^5 (x+3)^{-2}\,dx$

$= \left[\frac{(x+3)^{-1}}{1 \times -1}\right]_2^5$

$= \left[\frac{-1}{x+3}\right]_2^5$ **[1 mark]**

$= \frac{-1}{8} - \frac{-1}{5}$

$= \frac{3}{40}$ **[1 mark]**

3 $\frac{dy}{dx} = 8x - 7$

$y = 4x^2 - 7x + c$ **[1 mark]**

$y = 13$ when $x = -1$, so $13 = 4 \times (-1)^2 - 7 \times -1 + c$

$13 = 11 + c$

$c = 2$ **[1 mark]**

$y = 4x^2 - 7x + 2$ **[1 mark]**

4 Draw a sketch of $y = x^2 - 4x - 12$.

Identify the key features.

The graph is a parabola.

Let $y = 0$, $x^2 - 4x - 12 = 0$

$(x + 2)(x - 6) = 0$

$x = -2$ or 6 **[1 mark]**

Zeros are located at $(-2, 0)$ and $(6, 0)$.

The function has a minimum.

Find the derivative: $y' = 2x - 4$

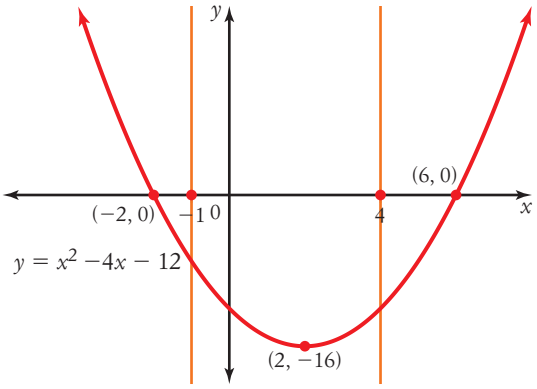
Let $y' = 0$: $0 = 2x - 4$

$x = 2$

When $x = 2$: $y = -16$ **[1 mark]**

Minimum at $(2, -16)$

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[1 mark]

$$\text{Required area} = \int_{-1}^{4}(x^2 - 4x - 12)dx$$

$$= \left[\frac{x^3}{3} - \frac{4x^2}{2} - 12x\right]_{-1}^{4}$$

$$= \left[\frac{x^3}{3} - 2x^2 - 12x\right]_{-1}^{4}$$ **[1 mark]**

$$= \left(\frac{4^3}{3} - 2 \times 4^2 - 12 \times 4\right) - \left(\frac{(-1)^3}{3} - 2 \times (-1)^2 - 12 \times -1\right)$$

$$= -\frac{176}{3} - \frac{29}{3}$$

$$= -\frac{205}{3}$$

$$= -68\frac{1}{3}$$

The negative sign means the area is below the x-axis.

Area $= 68\frac{1}{3}\text{units}^2$ **[1 mark]**

5 Solve simultaneous equations to find points of intersection of the curves.

$y = x^2$ (1)

$y = (x - 4)^2$ (2)

$x^2 = (x - 4)^2$

$x^2 = x^2 - 8x + 16$

$8x = 16$

$x = 2$ **[1 mark]**

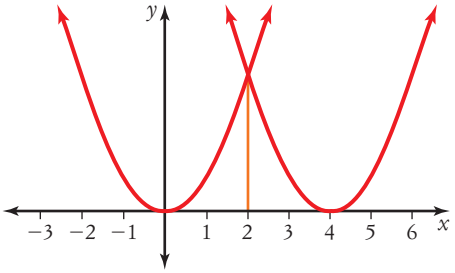
Both curves are parabolas.

The zero of $y = x^2$ is $(0, 0)$. **[1 mark]**

The zero of $y = (4 - x)^2$ is $(4, 0)$. **[1 mark]**

Draw a rough sketch.

Shade in the required area.



[1 mark]

$$\text{Area} = \int_0^2 x^2\,dx + \int_2^4 (x-4)^2\,dx$$

$$= \left[\frac{x^3}{3}\right]_0^2 + \left[\frac{(x-4)^3}{3}\right]_2^4$$ **[1 mark]**

$$= \left(\frac{8}{3} - \frac{0}{3}\right) + \left(\frac{0}{3} - \frac{-8}{3}\right)$$

$$= 5\frac{1}{3} \text{ units}^2$$ **[1 mark]**

6 $\text{Total change} = \int_0^3 F'(t)\,dt$ **[1 mark]**

$$= \int_0^3 100e^{0.2t}\,dt$$

$$= \left[\frac{100e^{0.2t}}{0.2}\right]_0^3$$

$$= \left[500e^{0.2t}\right]_0^3$$ **[1 mark]**

$$= 500e^{0.6} - 500e^0$$

$$= 411.05\ldots$$

In the first 3 hours, about 411 L flowed into the tank. **[1 mark]**

Part C – Analysis questions

1 $\text{Cross-sectional area} = \int_{-12}^{12} (15 - 0.4x^2)\,dx$ **[1 mark]**

$$= \left[15x - \frac{0.4x^3}{3}\right]_{-12}^{12}$$

$$= \left(15 \times 12 - \frac{0.4 \times 12^3}{3}\right) - \left(15 \times -12 - \frac{0.4 \times (-12)^3}{3}\right)$$

$$= -50.4 - 50.4$$

$$= -100.8$$ **[1 mark]**

Cross-sectional area $= 100.8 \text{ m}^2$

Volume = Area × Length

$$= 100.8 \times 80$$

$$= 8064 \text{ m}^3$$ **[1 mark]**

2 Let $A = (X, 0)$

Shaded area $= \int_0^X 3x^{\frac{1}{2}}\,dx$ **[1 mark]**

$$= \left[\frac{3x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^X$$

$$= \left[2x^{\frac{3}{2}}\right]_0^X$$

$$= 2X^{\frac{3}{2}} - 0$$

$$= 2X^{\frac{3}{2}}$$ **[1 mark]**

If $A = (X, 0), B = (X, 3X^{\frac{3}{2}})$ **[1 mark]**

Area of $OABC = l \times w$

$$= X \times 3X^{\frac{1}{2}}$$

$$= 3X^{\frac{3}{2}}$$ **[1 mark]**

$$\frac{\text{Shaded area}}{\text{Area of rectangle}} = \frac{2X^{\frac{3}{2}}}{3X^{\frac{3}{2}}}$$

Shaded area $= \frac{2}{3}$ area of rectangle. **[1 mark]**

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